(i)

$$\sum_{i=1}^{n} P(X=i) = 1$$

$$\implies \sum_{i=1}^{n} \left(\frac{1}{n} + \varepsilon_i\right) = 1$$

$$\implies \sum_{i=1}^{n} \varepsilon_i = 0$$

$$\begin{aligned} P(\text{same score on two rolls}) &= \sum_{i=1}^{n} \left(\frac{1}{n} + \varepsilon_i\right)^2 \\ &= \sum_{i=1}^{n} \left(\frac{1}{n^2} + \frac{2}{n}\varepsilon_i + \varepsilon_i^2\right) \\ &= \frac{1}{n} + \frac{2}{n}\sum_{i=1}^{n} \varepsilon_i + \sum_{i=1}^{n} \varepsilon_i^2 \\ &= \frac{1}{n} + \sum_{i=1}^{n} \varepsilon_i^2 \end{aligned}$$

Not all ε_i are 0, $\implies \sum_{i=1}^{n} \varepsilon_i^2 > 0$

$$P(\text{same score on two rolls}) > \frac{1}{n}$$

For a fair die,

$$P(\text{same score on two rolls}) = \frac{1}{n}$$

$\therefore$ Biased die is more likely

(ii) Let $S = \sum_{i=1}^{n} x_i \quad p_i = \frac{x_i}{S}$

$$\sum_{i=1}^{n} p_i = 1$$

By **(i)**,

$$\sum_{i=1}^{n} p_i^2 \geq \frac{1}{n}$$

$$\sum_{i=1}^{n} \left(\frac{x_i}{S}\right)^2 \geq \frac{1}{n}$$

$$\sum_{i=1}^{n} x_i^2 \geq \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2$$

$$\left(\sum_{i=1}^{n} x_i\right)^2 = \sum_{i=1}^{n} x_i^2 + 2\sum_{i=2}^{n}\sum_{j=1}^{i-1} x_i x_j$$

$$\begin{aligned} 2\sum_{i=2}^{n}\sum_{j=1}^{i-1} x_i x_j &= \left(\sum_{i=1}^{n} x_i\right)^2 - \sum_{i=1}^{n} x_i^2 \\ &\leq \left(\sum_{i=1}^{n} x_i\right)^2 - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2 \\ &= \frac{n-1}{n}\left(\sum_{i=1}^{n} x_i\right)^2 \end{aligned}$$

$$\implies \sum_{i=2}^{n}\sum_{j=1}^{i-1} x_i x_j \leq \frac{n-1}{2n}\left(\sum_{i=1}^{n} x_i\right)^2$$

(iii)

$$\begin{aligned} P(\text{biased}) &= \sum_{i=1}^{n}\left(\frac{1}{n} + \varepsilon_i\right)^3 \\ &= \sum_{i=1}^{n}\left(\frac{1}{n^3} + \frac{3\varepsilon_i}{n^2} + \frac{3\varepsilon_i^2}{n} + \varepsilon_i^3\right) \end{aligned}$$

$$P(\text{fair}) = n\left(\frac{1}{n}\right)^3 = \frac{1}{n^2}$$

$$\begin{aligned} P(\text{biased}) - P(\text{fair}) &= \sum_{i=1}^{n}\frac{3\varepsilon_i}{n^2} + \sum_{i=1}^{n}\frac{3\varepsilon_i^2}{n} + \sum_{i=1}^{n}\varepsilon_i^3 \\ &= \sum_{i=1}^{n}\frac{3\varepsilon_i^2}{n} + \sum_{i=1}^{n}\varepsilon_i^3 \\ &= \sum_{i=1}^{n}\varepsilon_i^2\left(\frac{3}{n} + \varepsilon_i\right) \end{aligned}$$

$$\because \frac{1}{n} + \varepsilon_i \geq 0$$

$$\therefore \frac{3}{n} + \varepsilon_i \geq \frac{2}{n} > 0$$

Not all ε_i are 0,

$$\sum_{i=1}^{n} \varepsilon_i^2 \left(\frac{3}{n} + \varepsilon_i \right) > 0$$

$$P(\text{biased}) > P(\text{fair})$$

This solution was submitted by **Jon**
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Last updated: July 21, 2026